World2Data: Real-Time Robot Scene Understanding

Problem
Robots often get raw video but lack structured, action-ready understanding: what objects are present, where they are, how they can be used, and what changed over time. Bounding boxes alone are not enough for safe navigation and manipulation.

Approach
We built a unified perception pipeline and live app that fuses:
- Object detection (YOLO / RF-DETR)
- Segmentation (FastSAM / RF-DETR-Seg / SAM3)
- Tracking (stable object IDs over time)
- Hand-object interactions (MediaPipe)
- Vision-language reasoning (LFM) for state + affordance summaries

The system runs in live mode and post-analysis mode. After recording, it automatically processes frames and renders synchronized views: Original, Combined, Tracking, Detections, Segmentation, Interactions, VLM Output, Temporal Changes, Ground Truth JSON, and Accuracy.

Key Results
- Produces machine-usable outputs instead of only visuals: per-frame objects, masks, interactions, temporal events, and JSON artifacts.
- Improves robotic actionability: better grasp regions from masks, interaction cues from hand contact, and semantic hints from VLM (e.g., object state, possible actions).
- Gives engineering visibility with per-stage timing (detector, segmenter, interactions, VLM, frame total), making bottlenecks measurable and optimizable.

Impact
This system helps move robots from pixel perception to task-relevant understanding. It supports planning, imitation learning data creation, behavior debugging, and evaluation in one workflow. The main value is not a single model, but the integration: geometric perception + temporal consistency + semantic reasoning in a real-time, operator-friendly interface.

Tech Stack
Python, Streamlit, OpenCV, PyTorch, Ultralytics YOLO, RF-DETR(-Seg), FastSAM, SAM3, MediaPipe Tasks, and Liquid LFM (MLX/Transformers backends).